

Glossary

Advanced Data Analytics



Terms and definitions from Course 5

#

%%time: A magic command that provides the runtime of the cell it's entered in to

A

Accuracy: The proportion of data points that were correctly categorized

AdaBoost: (Refer to **adaptive boosting**)

Adaptive boosting: A boosting methodology where each consecutive base learner assigns greater weight to the observations incorrectly predicted by the preceding learner

Affinity: The metric used to calculate the distance between points/clusters

Agglomerative clustering: A clustering methodology that works by first assigning every point to its own cluster, then progressively combining clusters based on intercluster distance

Average: The distance between each cluster's centroid and other clusters' centroids

B

Bagging: A technique used by certain kinds of models that use ensembles of base learners to make predictions; refers to the combination of bootstrapping and aggregating

Base learner: Each individual model that comprises an ensemble

Black-box model: Any model whose predictions cannot be precisely explained

Boosting: A technique that builds an ensemble of weak learners sequentially, with each consecutive learner trying to correct the errors of the one that preceded it

Bootstrapping: Sampling with replacement

Bayes' Theorem: An equation that can be used to calculate the probability of an event, given another event

C

Categorical variables: Variables that contain a finite number of groups or categories

Centroid: The center of a cluster determined by the mathematical mean of all the points in that cluster

Child node: A node that is pointed to/split from another node

Class imbalance: When a dataset has a target variable that contains more instances of one outcome than another

Collaborative filtering: A technique used by recommendation systems to make comparisons based on who else liked/disliked the content

Complete linkage: In clustering models, when the distance between two clusters is defined as the maximum pairwise distance between them

Content-based filtering: A technique used by recommendation systems to make comparisons based on attributes of content

Continuous variables: Variables that can take on an infinite and uncountable set of values

Cross-validation: A process that uses different portions of the data to test and train a model on different iterations

Customer churn: The business term that describes how many and at what rate customers stop using a product or service, or stop doing business with a company

D

DBSCAN: A clustering methodology that searches data space for continuous regions of high density; stands for “density-based spatial clustering of applications with noise”

Decision node: A node of the tree where decisions are made

Decision tree: A flowchart-like structure that uses branching paths to predict the outcomes of events, or the probability of certain outcomes

Discrete features: Features with a countable number of values between any two values

distance_threshold: A hyperparameter in agglomerative clustering models that determines the distance above which clusters will not be merged

Documentation: An in-depth guide that is written by the developers who created a package that features very specific information on various functions and features

Downsampling: A technique to handle class imbalance involving removing some observations from the majority class, making it so they make up a smaller percentage of the dataset than before

E

Ensemble learning: Building multiple models and aggregating their predictions

Ensembling: (Refer to **ensemble learning**)

eps (Epsilon): In DBSCAN clustering models, a hyperparameter that determines the radius of a search area from any given point

Extrapolation: A model’s ability to predict new values that fall outside of the range of values in the training data

F

F1-score: The harmonic mean of precision and recall

Feature engineering: The process of using practical, statistical, and data science knowledge to select, transform, or extract characteristics, properties, and attributes from raw data

Feature extraction: A type of feature engineering that involves taking multiple features to create a new one that may improve the performance of the algorithm

Feature selection: A type of feature engineering that involves selecting the features in the data that contribute the most to predicting the response variable

Feature transformation: A type of feature engineering that involves modifying existing features in a way that may improve model performance

G

Gradient boosting: A boosting methodology where each base learner in the sequence is built to predict the residual errors of the model that preceded it

Gradient boosting machines (GBMs): Model ensembles that use gradient boosting

GridSearch: A tool used to confirm that a model achieves its intended purpose by systematically checking combinations of hyperparameters to identify which set produces the best results, based on the selected metric

H

Hyperparameter tuning: Changing parameters that directly affect how the model trains, before the learning process begins

Hyperparameters: Parameters that can be set by the modeler before the model is trained

I

Inertia: In clustering models, the sum of the squared distances between each observation and its nearest centroid

Integrated Development Environment (IDE): A piece of software that has an interface to write, run, and test a piece of code

K

K-means: An unsupervised partitioning algorithm used to organize unlabeled data into groups, or clusters

L

Leaf node: In tree-based models, the deepest-level nodes, where a final prediction is made

learning_rate: In XGBoost, a hyperparameter that specifies how much weight is given to each consecutive tree's prediction in the final ensemble

Linkage: In clustering models, the method used to determine how to merge clusters

M

Machine learning: The use and development of algorithms and statistical models to teach computer systems to analyze and discover patterns in data without being explicitly programmed to do so

Magic commands: Commands that are built into IPython to simplify common tasks; always begin with either “%” or “%%”

Magics: (Refer to **magic commands**)

max_depth: In tree-based models, a hyperparameter that controls how deep each base learner tree will grow

max_features: In decision tree and random forest models, a hyperparameter that specifies the number of features that each tree randomly selects during training called “colsample_bytree” in XGBoost

min_samples_leaf: In decision tree and random forest models, a hyperparameter that defines the minimum number of samples for a leaf node called “min_child_weight” in XGBoost

min_samples_split: In decision tree and random forest models, a hyperparameter that defines the minimum number of samples that a node must have to split into more nodes

min_child_weight: In XGBoost models, a hyperparameter indicating that a tree will not split a node if it results in any child node with less weight than this value called “min_samples_leaf” in decision tree and random forest models

min_samples: In DBSCAN clustering models, a hyperparameter that specifies the number of samples in an ϵ -neighborhood for a point to be considered a core point (including itself)

Model selection: The process of determining which model should be the final product and put into production

Model validation: The set of processes and activities intended to verify that models are performing as expected

N

n_clusters: In K-means and agglomerative clustering models, a hyperparameter that specifies the number of clusters in the final model

n_estimators: In random forest and XGBoost models, a hyperparameter that specifies the number of trees your model will build in its ensemble

Naive Bayes: A supervised classification technique that is based on Bayes's Theorem with an assumption of class-conditional independence among predictors

P

Plan stage: The part of the PACE workflow process where a data profession first starts thinking about what the problem actually is and what needs to be done to find a solution

Popularity bias: The phenomenon of more popular items being recommended too frequently

Posterior probability: The probability of an event occurring after taking into consideration new information

Precision: The proportion of positive predictions that were correct to all positive predictions

R

Random forest: An ensemble of decision trees trained on bootstrapped data with randomly selected features

Recall: The proportion of actual positives that were identified correctly to all actual positives

Recommendation systems: Unsupervised learning techniques that use unlabeled data to offer relevant suggestions to users

Root node: In tree-based models, the first node of the tree, where the first decision is made

S

Shrinkage: (Refer to `learning_rate`)

Silhouette analysis: The comparison of different models' silhouette scores

Silhouette score: The mean of the silhouette coefficients of all the observations in a model

Single linkage: In clustering models, when the distance between two clusters is defined as the minimum pairwise distance between clusters

Supervised learning: A category of machine learning that uses labeled datasets to train algorithms to classify or predict outcomes

T

Tree-based learning: A type of supervised machine learning that uses decision trees to perform classification and regression tasks

U

Unsupervised learning: A category of machine learning that uses algorithms to analyze and find structure in unstructured or unlabeled data

Upsampling: A technique to handle class imbalance that involves taking observations from the minority class and either adding copies of those observations to the dataset or generating new observations to add to the dataset so they make up a larger percentage of the dataset than before

W

Ward linkage: In clustering models, a method that merges two clusters whose merging will result in the lowest inertia

Weak learner: A model that performs slightly better than randomly guessing

X

XGBoost (extreme gradient boosting): An optimized GBM package

Z

Zero Frequency problem: In Naive Bayes models, a problem that occurs when the dataset has no occurrences of a class label and some value of a predictor variable together, which would cause a predicted probability of 0